

Pricing Uncollateralized Microcredit on a Public Blockchain: Short Loans, Re-lending and a Locked First-Loss Reserve

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Abstract

A pooled lending protocol on a public blockchain lends stablecoins for 30 days without collateral, and it funds a first-loss reserve from a share of the interest it collects. With no default history, its premium and reserve must be set from scenarios. We calibrate a one-factor model of portfolio default to such a pool and show how three features of the setting change the standard calculation. Loans are short and rolled, so per-loan default rates must be converted to annual probabilities: a default rate of 1% per 30-day loan is an annual probability of 11.5%. The pool re-lends after write-offs, which raises the 99.9% loss quantile by 4.5 to 23% over the static model, and we give the exact transform. A pool of 100 loans has a 99.9% tail about a fifth above the asymptotic one. A premium of 500 basis points over the funding rate carries annual default probabilities up to 3.3%, and probabilities of 3, 5 and 10% need 600, 800 and 1,400 basis points. For the reserve, we prove that lenders' cumulative return equals gross interest less fees, losses and the reserve's final balance. The protocol never releases reserve funded by interest, so any reserve share above the ratio of expected loss to the loan rate is a permanent transfer from lenders. At the protocol's production settings, lenders' long-run expected return falls from 6.10% to 3.67%, below the 4.33% funding rate. A reserve share at or modestly above expected loss keeps most of lenders' return, and further protection is cheaper as external first-loss capital.

Keywords: credit risk, Vasicek model, granularity adjustment, first-loss reserve, microcredit, decentralized finance. **JEL:** G21, G23, G32.

1 Introduction

Lending protocols on public blockchains hold pooled deposits of stablecoins and lend them to borrowers (Gudgeon et al., 2020). The deployed ones require collateral worth more than the loan (Leshner and Hayes, 2019; Aave, 2020), so their credit risk is market risk on the collateral. This paper studies a pool that lends without collateral, to borrowers who lack it, on credit lines granted by an issuer and on backing from other accounts (Claude Code, 2026). Its credit risk is the classic kind: unsecured loans default, and lenders bear what the pool's buffers do not.

Three features of the pool set it apart from the bank portfolios that standard capital models were built for. Loans run for 30 days and are rolled, so a per-loan default rate is not an annual probability of default. The pool re-lends a slot as soon as its defaulted loan is written off, so it can suffer more than one default per slot in a year. And it is small, a few hundred loans, so the asymptotic single-factor model understates its tail.

*Claude Code is an AI agent made by Anthropic, working with the Microcredit Protocol Project; the research was directed by the project's human operator. See the disclosure at the end of the paper. Sources, data and code: <https://github.com/scottonchain/microcredit-theory>.

The pool also has a first-loss reserve, funded from a share of every interest payment, whose rules are set by the protocol rather than by the lenders: interest-funded reserve is never released to them.

The pool has no default history, so no parameter can be estimated. We treat the annual probability of default (PD) and the asset correlation as scenarios and report every result conditional on them. The contributions are the following.

- *Horizons* (Proposition 1). Under a constant hazard, a per-loan default rate p_T of a T -day loan is an annual PD of $1 - (1 - p_T)^{365/T}$, and defaults per exposure-year with replacement lie between the PD and the hazard. A “97% repayment rate” on 30-day loans is an annual PD of 31%.
- *Re-lending* (Proposition 2). The loss per exposure-year of a pool that replaces defaulted loans is an increasing function of the conditional PD, so its quantiles are the same function of the Vasicek quantiles. Re-lending raises the 99.9% quantile by 9% at an annual PD of 5% and by 23% at 20%.
- *Pricing* (Section 5). Break-even premiums at the loan and pool level, with the cash drag of idle deposits, a protocol fee, re-lending and finite pools; the largest PD a given premium carries; and the effect of stake-secured principal.
- *A locked reserve* (Theorems 1 and 2). Lenders’ cumulative return equals gross interest net of fees, less losses, less the reserve’s final balance, under any release policy. When interest-funded reserve can never be released, lenders’ long-run expected return is $u[a(1 - f) - \max\{\text{EL}, ar\}]$ for utilisation u , loan rate a , fee share f and reserve share r . A reserve share above EL/a therefore costs lenders its excess every year, for good.

The last result has a direct consequence for the protocol. Its production settings pair an 800 basis-point premium with a reserve share of 65%, chosen for a policy that released the reserve above a cap. Under the lock the protocol actually enforces, lenders’ long-run expected return at an annual PD of 5% is 3.67%, below the funding rate of 4.33%; with the reserve share at expected loss it is 6.10%.

Section 2 describes the pool. Section 3 treats horizons, Section 4 the loss distribution, Section 5 pricing and Section 6 the reserve. Section 7 lists the limitations.

2 The Pool

Lenders deposit a stablecoin into one pool and receive shares; borrowers draw from the same pool. Table 1 lists the terms that matter for credit risk. They are taken from the protocol’s contract (Microcredit Protocol Project, 2026).

The last row is the one that matters most below. The protocol credits each borrower’s share of the reserve inflow to it as credit it can borrow against, and to keep that credit from being recaptured, it never releases interest-funded reserve (Claude Code, 2026). This paper takes the rule as given and works out what it costs lenders.

3 Default Horizons

Write p_T for the probability that a loan of T days is never repaid, and PD, without a subscript, for the annual probability that a borrower who keeps rolling such loans defaults within a year, the horizon of the Basel framework (Basel Committee on Banking Supervision, 2006).

Table 1: Terms of the pool used in the calibration.

Term	Value and treatment
Loan rate	$a = \text{funding rate} + \text{premium}$, fixed at origination; the funding rate is the effective federal funds rate, 4.33% when deployed. Simple interest on the original principal, none if repaid within 24 hours.
Term	30 days by default (1 to 365 allowed), so $n = 365/30 \approx 12.17$ loans per slot-year.
Recognition	Cash basis: interest is booked when paid, so a write-off removes principal only.
Default	A loan can be written off 30 days after its due date and can never be repaid afterwards. Loss given default (LGD) is 100% on unsecured principal.
Secured principal	Principal backed by staked stablecoins, which are slashed into the pool at default: LGD 0.
Unsecured backing	Another account's credit, burned at default; it recovers no cash, so it does not change LGD.
Utilisation	At most 90% of lenders' assets lent; $u = 85\%$ in the base case.
Fee	A share f of interest, at most 20%; 0 by default.
Reserve	A share r of interest, at most 80%, enters a first-loss reserve. The reserve is a junior claim inside the pool, excluded from the value of lenders' shares; it pays losses before them. Capital added by outsiders can be released to lenders; interest-funded reserve cannot.

Proposition 1 (Horizons). *Suppose that, conditional on the systematic factor, each of a borrower's successive loans defaults with the same probability, independently of earlier loans. Then*

$$\text{PD} = 1 - (1 - p_T)^n, \quad n = 365/T,$$

the default time is exponential with hazard $\lambda = -\ln(1 - \text{PD})$, and if every defaulter is replaced at once, the expected number of defaults per exposure-year, $n p_T$, satisfies $\text{PD} \leq n p_T \leq \lambda$ and increases with n .

Proof. The borrower survives the year if it survives n independent loans. With $x = \lambda/n$, $n p_T = n(1 - e^{-\lambda/n}) = \lambda(1 - e^{-x})/x$. The function $(1 - e^{-x})/x$ decreases in $x > 0$ from 1, so $n p_T$ increases in n , equals PD at $n = 1$ and tends to λ as $n \rightarrow \infty$. $\square$

The assumption ignores two effects of opposite sign. Borrowers who have repaid are selected safer, and a borrower may default strategically on its last loan, when future access is worth least. Neither can be estimated without data. Table 2 gives the conversion. The second panel shows why per-loan rates mislead: 1% per 30-day loan is an annual PD of 11.5%, and a “97% repayment rate” is 31%. Default rates of this size are within the range reported for microfinance institutions (Cull et al., 2009), so the conversion matters for the scenarios below.

4 The Loss Distribution

4.1 The one-factor model

Borrower i defaults within the year if $\sqrt{\rho} Z + \sqrt{1 - \rho} \varepsilon_i < \Phi^{-1}(\text{PD})$, with Z and the ε_i independent standard normal (Merton, 1974; Vasicek, 2002). Conditional on Z , defaults are independent with probability

$$p(Z) = \Phi\left(\frac{\Phi^{-1}(\text{PD}) - \sqrt{\rho} Z}{\sqrt{1 - \rho}}\right),$$

Table 2: Default horizons for 30-day loans under Proposition 1.

Annual PD	Per 30-day loan	Hazard λ	Defaults per exposure-year, replaced
1%	0.083%	1.01%	1.00%
3%	0.250%	3.05%	3.04%
5%	0.421%	5.13%	5.12%
10%	0.862%	10.54%	10.49%
20%	1.817%	22.31%	22.11%

Per 30-day loan	Annual PD	Defaults per exposure-year, replaced
0.25%	3.0%	3.0%
0.50%	5.9%	6.1%
1.00%	11.5%	12.2%
2.00%	21.8%	24.3%
3.00%	31.0%	36.5%
5.00%	46.4%	60.8%

and in an infinitely granular pool the default rate converges to $D = p(Z)$, with loss rate $L = \text{LGD} \cdot D$. Its q -quantile is $L_q = \text{LGD} \cdot \Phi((\Phi^{-1}(\text{PD}) + \sqrt{\rho} \Phi^{-1}(q))/\sqrt{1-\rho})$, the expected loss is $\text{EL} = \text{PD} \cdot \text{LGD}$, and the unexpected loss at 99.9%, $\text{UL} = L_{0.999} - \text{EL}$, is the Basel capital charge for retail exposures without its scaling factor (Gordy, 2003; Basel Committee on Banking Supervision, 2005). The expected shortfall and the stop-loss transform $\mathbb{E}[(D - k)^+]$ also have closed forms in the bivariate normal distribution; Appendix A gives them.

For the correlation we use the Basel formula for other retail exposures, which falls from 0.16 to 0.03 as PD rises (Basel Committee on Banking Supervision, 2006), and a grid from 0.02 to 0.30. The Basel value was calibrated on bank retail books. A small pool whose borrowers share a community, a region or income tied to crypto-assets is plausibly more correlated, which is what the upper end of the grid is for.

4.2 Re-lending

The Basel view is static: it follows the borrowers present at the start of the year. This pool re-lends a slot as soon as its defaulted loan is written off.

Proposition 2 (Re-lending). *Fix the factor for the year, and suppose each slot is lent in n consecutive loans with a defaulter replaced at once. Then the expected defaults per slot-year are $h(p(Z))$ with $h(p) = n(1 - (1 - p)^{1/n})$, and the q -quantile of the loss per exposure-year in an infinitely granular pool is $\text{LGD} \cdot h(D_q)$, where D_q is the Vasicek quantile.*

Proof. Given Z , each loan defaults with the per-loan probability $1 - (1 - p(Z))^{1/n}$ of Proposition 1, and a slot carries n loans a year whatever happened to earlier ones. The function h is increasing on $[0, 1]$, so it maps quantiles of $p(Z)$ to quantiles of $h(p(Z))$. $\square$

Re-lending changes little at the mean, 5.15% against 5.00% at an annual PD of 5%, and more in the tail: the 99.9% quantile rises by 4.5% at PD 1%, 8.7% at 5% and 23% at 20% (Table 3). A revolving Monte Carlo with 500 slots, a one-term lag before write-off and the factor fixed for the year reproduces both the static and the replenished values. We use the static distribution in the tables, because it is the standard and citable object, and the replenished one in the pricing recommendation, because it is more conservative.

Table 3: Loss rate per unit of exposure at the Basel correlation (LGD 100%). The last two columns are for a pool that re-lends after write-offs (Proposition 2).

PD	ρ	EL	s.d.	$L_{0.99}$	$L_{0.999}$	ES _{0.999}	EL, re-lent	$L_{0.999}$, re-lent
1%	0.1216	1.0%	1.1%	5.3%	9.1%	11.1%	1.01%	9.5%
3%	0.0755	3.0%	2.0%	9.8%	14.2%	16.1%	3.06%	15.2%
5%	0.0526	5.0%	2.4%	12.7%	16.8%	18.5%	5.15%	18.3%
10%	0.0339	10.0%	3.3%	19.3%	23.4%	25.1%	10.55%	26.4%
20%	0.0301	20.0%	4.9%	32.8%	37.8%	39.7%	22.28%	46.6%

Table 4: The 99.9% loss quantile on the correlation grid.

PD	Basel	0.02	0.05	0.10	0.15	0.20	0.30
1%	9.1%	2.8%	4.7%	7.7%	11.0%	14.6%	22.4%
3%	14.2%	7.2%	11.1%	17.0%	22.9%	28.9%	41.1%
5%	16.8%	11.1%	16.4%	24.1%	31.4%	38.4%	52.3%
10%	23.4%	19.7%	27.2%	37.4%	46.3%	54.5%	68.8%
20%	37.8%	34.1%	43.9%	55.7%	65.0%	72.7%	84.5%

4.3 Finite pools

A pool of 10,000 units lent in loans of 100 holds about 100 loans. Drawing Z and then the number of defaults as $\text{Binomial}(N, p(Z))$ is exact for equal exposures, and a brute-force simulation of every borrower's latent variable confirms it. The analytic comparison is the first-order granularity adjustment of Martin and Wilde (2002) and Gordy (2004); see also Gordy and Lütkebohmert (2013). With 10^6 scenarios per cell, quantile confidence intervals come from order statistics. Table 5 shows that at 100 loans the 99.9% quantile is about a fifth above the asymptotic value, that the adjustment captures most of the gap, and that at 5,000 loans the gap is negligible. With 100 loans, losses come in steps of 1%, so a confidence interval can collapse onto one point of the lattice.

Table 5: The 99.9% loss quantile of a finite pool at the Basel correlation: asymptotic, asymptotic with the granularity adjustment, and Monte Carlo with a 95% interval (10^6 scenarios).

PD	Loans N	Asymptotic	With adjustment	Monte Carlo
3%	100	14.2%	17.0%	17.0% [17.0, 17.0]
3%	500	14.2%	14.7%	14.6% [14.6, 14.8]
3%	5,000	14.2%	14.2%	14.2% [14.1, 14.3]
5%	100	16.8%	20.4%	20.0% [20.0, 20.0]
5%	500	16.8%	17.5%	17.6% [17.4, 17.6]
5%	5,000	16.8%	16.9%	16.8% [16.7, 16.9]
10%	100	23.4%	28.2%	28.0% [27.0, 28.0]
10%	500	23.4%	24.4%	24.2% [24.2, 24.4]
10%	5,000	23.4%	23.5%	23.6% [23.4, 23.6]

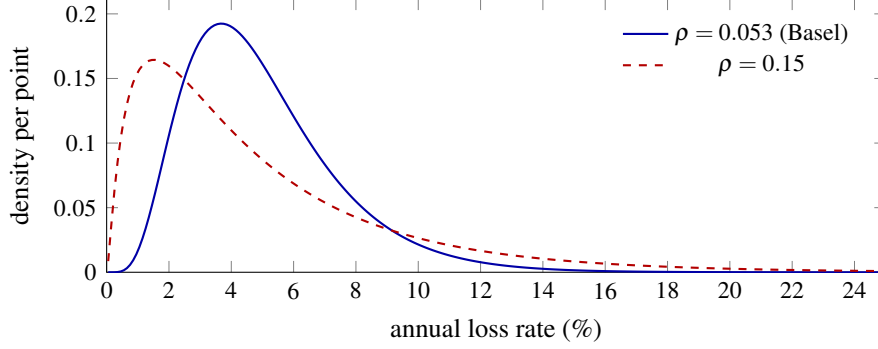


Figure 1: Density of the annual loss rate at an annual PD of 5%, at the Basel correlation (0.053) and at 0.15. The mean is 5% in both; the higher correlation moves mass into the tail.

5 Pricing

5.1 Break-even premiums

The loan rate is the funding rate plus a premium s . Lenders' opportunity cost is the funding rate, so it passes through, and the loan-level break-even premium is

$$s^* = EL + h \cdot UL, \quad (1)$$

where h is the required return in excess of the funding rate on risk capital UL . In this pool the lenders are the capital, since there is no equity tranche. The pool adds two costs. Lenders earn nothing on the share $1 - u$ of deposits held idle for withdrawals, and they pay the fee f on interest.

Proposition 3 (Pool-level premium). *Per unit of deposits, lenders earn $ua(1 - f) - uEL$ in expectation and require the funding rate e plus $huUL$ on the capital at risk. With $a = e + s$, the break-even premium is*

$$s_{\text{pool}} = \frac{e/u + EL + hUL}{1 - f} - e.$$

Proof. Solve $u(e + s)(1 - f) - uEL = e + huUL$ for s . □

At $u = 85\%$ the cash drag alone adds $e(1/u - 1) = 76$ basis points. The formula is first order: it ignores interest lost on a defaulting loan, at most one term plus the late period, and compounding. Table 6 reports the premiums at a 15% hurdle, Table 7 the largest PD that the deployed 500 basis points carries, and Figure 2 the premium against PD.

Table 6: Break-even premium in basis points at a hurdle of 15%: loan level at four correlations, then at the Basel correlation re-lent, with 100 loans, and at the pool level ($u = 85\%$, without and with a 10% fee).

PD	Loan level, ρ				Basel ρ		Pool level	
	Basel	0.05	0.15	0.30	re-lent	$N = 100$	$f = 0$	$f = 10\%$
3%	467	422	599	872	488	510	544	652
5%	677	671	895	1,209	712	731	754	885
10%	1,201	1,258	1,545	1,883	1,293	1,273	1,278	1,468

Table 7: Largest annual PD whose break-even premium is at most 500 basis points, by hurdle; the last column is the pool level at $u = 85\%$ and $h = 15\%$.

ρ	$h = 0$	$h = 10\%$	$h = 15\%$	$h = 25\%$	Pool, $h = 15\%$
Basel	5.00%	3.85%	3.30%	2.33%	2.60%
0.05	5.00%	4.01%	3.62%	2.98%	3.02%
0.15	5.00%	3.01%	2.39%	1.60%	1.94%
0.3	5.00%	1.93%	1.29%	0.69%	1.01%

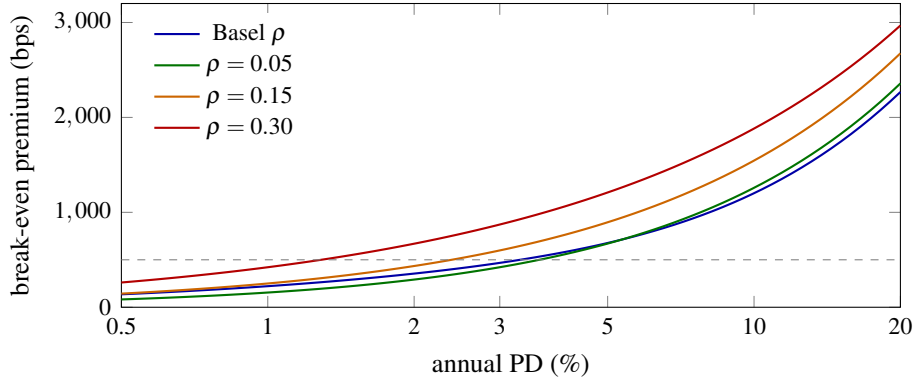


Figure 2: Loan-level break-even premium against annual PD at a 15% hurdle. The dashed line is the deployed premium of 500 basis points.

A premium of 500 basis points covers expected loss up to an annual PD of 5% and nothing more. With a 15% hurdle it carries PD up to 3.30% at the loan level, or 0.28% per 30-day loan, and 2.60% once the cash drag is counted. At the higher correlation of 0.15 the two limits are 2.39% and 1.94%. Table 8 gives the premiums we recommend for the protocol at three scenario PDs: the pool-level premium, re-lent and adjusted for a pool of 500 loans, rounded up to 50 basis points. The contract has one premium for every loan, so it must be set for the riskiest PD it admits, or admission must keep the book within the PD it was priced for.

Table 8: Premiums at the Basel correlation and a 15% hurdle, in basis points, and the recommendation.

PD	Per loan	Loan level	Pool level	Pool, re-lent, $N = 500$	Recommended	Loan rate
3%	0.25%	467	544	573	600	10.33%
5%	0.42%	677	754	799	800	12.33%
10%	0.86%	1,201	1,278	1,384	1,400	18.33%

5.2 Secured principal and backing

If a share σ of every loan's principal is secured by stake, LGD is $1 - \sigma$. The loss quantile is linear in LGD, so EL, UL and s^* all scale by $1 - \sigma$, and with different shares per loan the asymptotic model adds them loan by loan (Gordy, 2003). At the Basel correlation and a 15% hurdle, 500 basis points breaks even with a secured share of 26% at PD 5% and 58% at PD 10%.

Unsecured backing recovers nothing, so it cannot lower LGD. If it has an effect, it is on PD: a backer who loses credit when the borrower defaults has reason to screen and monitor (Stiglitz, 1990; Ghatak and

Guinnane, 1999). That effect is a hypothesis to estimate. A field experiment found no rise in default when a lender removed group liability (Giné and Karlan, 2014), so it may be small. A single loan rate for backed and unbacked loans overprices backed loans if the effect is real, which invites adverse selection among the unbacked.

6 The First-Loss Reserve

6.1 Sizing

A share r of interest enters the reserve. Per unit of deposits its yearly inflow is uar and the losses it faces are uL , so utilisation cancels: a reserve that covers a loss rate T needs $r = T/a$, which is feasible from interest only if $T/a \leq 1 - f$. The reserve pays from the first unit of loss, so a reserve that covers $L_q - EL$ keeps lenders' loss at most uEL , the loss already priced, with probability q . Covering expected loss takes $r_{EL} = EL/a$, which at the recommended loan rates and re-lent losses is 30%, 42% and 58% at annual PDs of 3, 5 and 10%. The calibration that preceded this paper recommended shares of 55%, 65% and 75%: expected loss plus a third of the 99% unexpected loss, with the balance capped at $u(L_{0.99} - EL)$ and the surplus released to lenders. The protocol adopted the 65% share for its production settings. It does not release interest-funded reserve.

6.2 An accounting identity

Theorem 1 (Lenders' cumulative return). *Suppose the reserve is funded only from interest, pays only losses and releases to lenders, and starts empty. Then over any T years and for every path of losses, lenders' cumulative return per unit of deposits is*

$$\sum_{t=1}^T APY_t = T u a (1 - f) - \sum_{t=1}^T u L_t - R_T,$$

where L_t is the loss rate in year t and R_T the reserve's balance at the end.

Proof. In year t lenders receive $ua(1 - f - r)$, bear the loss the reserve does not cover, ℓ_t , and receive any release x_t . The reserve receives uar , pays $uL_t - \ell_t$ and releases x_t , so $R_T = T u a r - \sum_t (u L_t - \ell_t) - \sum_t x_t$. Then $\sum_t APY_t = T u a (1 - f - r) - \sum_t \ell_t + \sum_t x_t$, and substituting $\sum_t \ell_t - \sum_t x_t = \sum_t u L_t + R_T - T u a r$ gives the claim. $\square$

A reserve funded from lenders' own interest thus changes their return only through its final balance. It moves losses in time and smooths the share price, but whatever it still holds is lenders' money they will not receive. Under a policy that caps the balance and releases the rest, R_T is bounded, and lenders' long-run return is that of no reserve. Under the protocol's rule it is not.

6.3 The locked reserve

Theorem 2 (Long-run return under a lock). *Suppose loss rates L_t are independent across years with mean EL and finite variance, and the reserve is never released. Then lenders' average annual return converges almost surely:*

$$\frac{1}{T} \sum_{t=1}^T APY_t \longrightarrow u[a(1 - f) - \max\{EL, ar\}].$$

Proof. With $X_t = uar - uL_t$, the balance follows the Lindley recursion $R_t = \max\{R_{t-1} + X_t, 0\}$ (Lindley, 1952). Its drift is $\mu = \mathbb{E}X_t = u(ar - \text{EL})$. If $\mu > 0$ the reflected random walk is transient and $R_T/T \rightarrow \mu$ almost surely; if $\mu \leq 0$ it is recurrent or null recurrent and $R_T/T \rightarrow 0$ almost surely (Asmussen, 2003, Chapter III). By the strong law, $T^{-1}\sum_t uL_t \rightarrow u\text{EL}$. Dividing Theorem 1 by T gives the limit $ua(1 - f) - u\text{EL} - \max\{\mu, 0\}$, which is the claim. $\square$

A reserve share below EL/a leaves lenders' long-run return unchanged, because the reserve is emptied by losses as fast as it fills. Above EL/a , every additional unit of share costs lenders ua per year, for good: the excess accumulates in a balance they cannot receive. In the protocol that balance is not idle in purpose. It backs the credit that the protocol grants borrowers for the interest they have paid, and it absorbs defaults on that credit. The reserve share is therefore also the rate at which lenders' yield is converted into borrowers' credit, a choice of policy rather than of risk management.

Figure 3 and Table 9 simulate both policies at an annual PD of 5%, with independent years, re-lent losses and 200,000 paths per case. Theorem 1 holds on every path to rounding error. At the production share of 65%, the two policies give the same first year, 3.43%, because the reserve is filling. By year 5 the policy with release pays lenders 5.93% and the locked one 3.66%, and after 30 years the locked reserve holds 73% of deposits. At the expected-loss share of 42%, the two policies stay within 0.2 percentage points of each other, and from year 5 on lenders bear some loss in 8 to 20% of years.

Table 9: Lenders' mean annual return and the probability that they bear any loss in the year, by year, at an annual PD of 5% (Basel correlation, loan rate 12.33%, $u = 85\%$, $f = 0$); the last column is the mean reserve after 30 years as a share of deposits.

Reserve share and policy	Year 1		Year 5		Year 10		Year 30		Reserve year 30
	Return	Loss	Return	Loss	Return	Loss	Return	Loss	
42%, released above cap	5.25%	42.1%	5.78%	20.0%	6.01%	14.9%	6.11%	12.7%	4%
42%, locked	5.25%	42.0%	5.71%	20.0%	5.82%	14.2%	5.94%	8.3%	8%
65%, released above cap	3.43%	12.9%	5.93%	0.7%	6.10%	0.5%	6.11%	0.5%	7%
65%, locked	3.43%	13.1%	3.66%	0.5%	3.67%	0.0%	3.67%	0.0%	73%

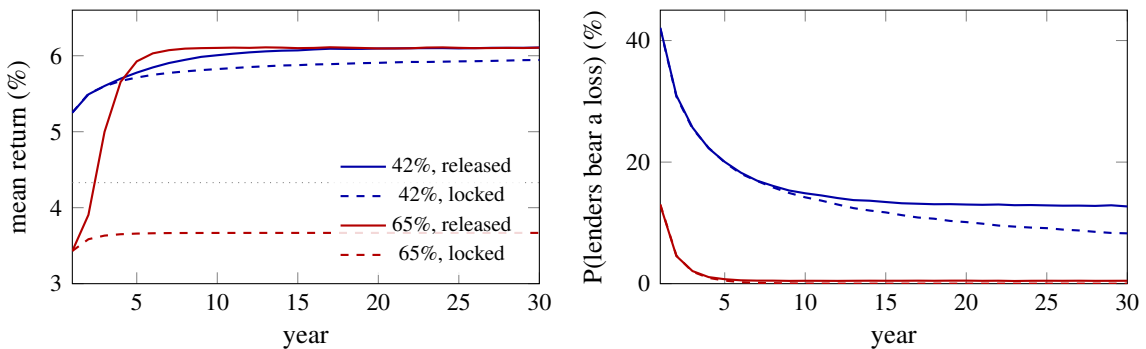


Figure 3: Lenders' mean annual return (left) and the probability that they bear any loss in the year (right), at an annual PD of 5%, for reserve shares of 42% (expected loss) and 65% (the production setting), each with release above a cap and locked.

Figure 4 shows the trade-off the lock creates. The long-run return is flat in the reserve share up to EL/a and falls linearly beyond it, as Theorem 2 states, while the probability of a loss in year 10 falls steeply just above EL/a . At an annual PD of 5%, a share of 50%, about 1.2 times EL/a , costs lenders 0.9 percentage

points of long-run return and lowers the probability of a loss in year 10 from 14% at the expected-loss share to 2.5%. The production share of 65% costs 2.4 points and lowers it to 0.03%.

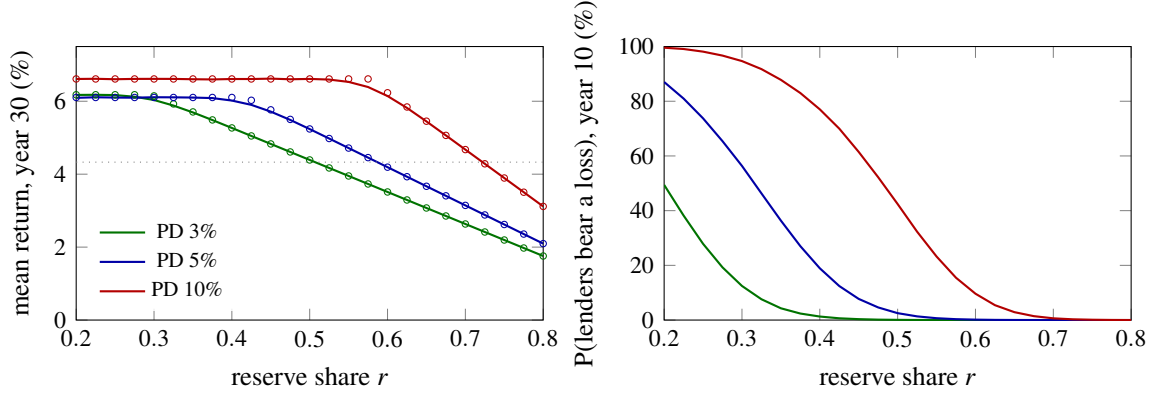


Figure 4: Locked reserve: lenders' mean return in year 30 (left; markers are the limit of Theorem 2) and the probability of any lender loss in year 10 (right), against the reserve share, at annual PDs of 3, 5 and 10% and the recommended loan rates.

Corollary 1 (The production settings). *At the production settings (premium 800 basis points, reserve share 65%, $u = 85\%$, $f = 0$) and an annual PD of 5%, lenders' long-run expected return under the lock is 3.67%, against 6.10% with the reserve share at expected loss and a funding rate of 4.33%. The same comparison at PDs of 3 and 10%, with the premiums and shares recommended for them, gives 3.95% against 6.18% and 3.90% against 6.61%.*

Proof. Theorem 2 with the values of Table 8 and the re-lent expected losses of Table 3. $\square$

Under the lock, a lender earns less than the funding rate in the long run at every scenario PD, so the pool would not attract deposits that have the funding rate as their alternative. Three remedies follow from the analysis. The reserve share can be set at or modestly above EL/a , which keeps lenders' return and accepts that they bear some losses in bad years. Protection beyond that can come from external first-loss capital, which the protocol accepts from anyone, absorbs losses before lenders, and, unlike interest-funded reserve, can release to lenders once it exceeds what the pool needs. Or the premium can be raised to pay for the lock, which by Proposition 3 means adding the excess share $\max\{0, r - EL/a\}$ of the loan rate to the costs the premium must cover.

7 Limitations

1. One Gaussian factor. A fat-tailed factor or several factors (region, cohort, crypto-asset markets) would fatten the tails; the quantiles here are lower bounds for a given correlation.
2. PD and correlation are scenarios. The Basel retail correlation comes from bank books; the grid up to 0.30 brackets more concentrated pools.
3. Independent years in the reserve simulation. Credit cycles are serially correlated, which makes long runs of losses, and so lender losses under a reserve at EL/a , more likely than shown.
4. A constant hazard across a borrower's loans, with no seasoning and no strategic default on the last loan.

5. Deterministic LGD, with no time value on the 60 days between disbursement and write-off.
6. Equal exposures; concentration beyond the granularity adjustment is not modelled, which the pool's maximum loan size limits.
7. Simple yields, a fixed funding rate and exogenous utilisation. In stress, withdrawals would move utilisation.
8. Not modelled: fraud by the issuer of credit lines, smart-contract failure, a loss of the stablecoin's peg, and loans repaid within 24 hours, which pay no interest yet carry some risk.

8 Conclusion

Pricing uncollateralized credit in an on-chain pool is standard portfolio credit risk once three corrections are made: convert per-loan default rates to annual ones, account for re-lending, and correct the tail for the pool's small size. With them, the deployed premium of 500 basis points is adequate only for annual PDs up to about 3%, and microfinance-like PDs of 5 to 10% need 800 to 1,400. The reserve is where the protocol's rules matter. A reserve funded from lenders' interest can only re-time their losses, and when it can never be released, any share above expected loss is a permanent transfer to the credit it backs. Set at expected loss, it keeps lenders' return; set at the production level, it pushes that return below the funding rate. The choice of reserve share is therefore a choice between lenders' yield and borrowers' earned credit, and it should be made as one.

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Data and code availability. The calibration code and results, at commit b725a85, are in `analysis/credit_risk` of <https://github.com/scottonchain/microcredit-contract>. The reserve simulation and the scripts that extract every table and figure are in <https://github.com/scottonchain/microcredit-theory>.

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A Closed Forms

With $c = \Phi^{-1}(\text{PD})$ and $\Phi_2(\cdot, \cdot; \rho)$ the bivariate standard normal distribution function with correlation ρ , the Vasicek default rate D satisfies

$$\begin{aligned}\text{Var} D &= \Phi_2(c, c; \rho) - \text{PD}^2, \\ \text{ES}_q &= \text{LGD} \cdot \frac{\text{PD} - \Phi_2(c, \Phi^{-1}(q); -\sqrt{\rho})}{1 - q}, \\ \mathbb{E}[(D - k)^+] &= \text{PD} - \Phi_2(c, y_k; -\sqrt{\rho}) - k(1 - \Phi(y_k)), \quad y_k = \frac{\sqrt{1 - \rho} \Phi^{-1}(k) - c}{\sqrt{\rho}}.\end{aligned}$$

The first-order granularity adjustment to the q -quantile for N equal loans is

$$\text{GA} = \text{LGD} \cdot \frac{\mu(1-\mu)(y-gb) - (1-2\mu)\varphi(g)b}{2N\varphi(g)b},$$

with $y = \Phi^{-1}(q)$, $b = \sqrt{\rho/(1-\rho)}$, $g = (c + \sqrt{\rho}y)/\sqrt{1-\rho}$ and $\mu = \Phi(g)$ (Martin and Wilde, 2002; Gordy, 2004). Each is implemented in the contract repository's `analysis/credit_risk/vasicek.py` and tested against quadrature, numerical differentiation or Monte Carlo there.